An orbit deficit for ambient-eligible tail cuts

Human Compression project: L044 working continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle$ be a transitive string permutation group generated by involutions. Write $G_j = \langle \rho_k : k \neq j \rangle$, and call a j-edge *ambient-eligible* when its endpoints lie in different G_j-orbits. Fix $0 \leq t < r-1$, put

$$I = \{t+1, \dots, r-1\}, \qquad H = G_{>t}, \qquad U = \mathrm{supp}(H), \qquad d = |I|.$$

Suppose there are at least two ambient-eligible edges of each colour in I. A tail selection Q chooses exactly two such edges per colour, on vertex set U. All actual edges of these colours lie in U and inside individual H-orbits. Full IP, evenness and connectedness of the noncommuting diagram are not assumed.

The paid paired-cut square-normal-form proof applies to this interval of colours: its orbit-cut exchanges use ambient eligibility, and every global middle-colour move has its middle index in I. Choose a square-only selection with minimum number q of squares. As in that proof, *every* such minimum selection has at most one square per component: otherwise minimize the distance between two squares over all minimum selections and apply the incident, ordinary or global middle move. Thus every component is a tree or is unicyclic with a square. No transitivity of H is used here.

Lemma 1 (Localisation and ordinary slides)**.** *If S is the square in a component C, every ambient-eligible edge of a square colour has both endpoints in C. An ordinary square slide along a selected bridge leaving S preserves the vertex set of C and all other component vertex sets.*

Proof. Let a be a square colour. An eligible a-edge cannot have both endpoints in another component: its selected connecting path would avoid a. If its endpoints belonged to different components, delete an a-edge of S and insert it. The deletion destroys S without disconnecting C, and the insertion joins components without creating a cycle. This contradicts minimum q and proves localisation.

For the slide write its first bridge as $v -_\ell w$, with $v \in S$ and ℓ commuting with a square colour a. Put $v' = v^{\rho_a}$ and $w' = w^{\rho_a} = (v')^{\rho_\ell}$. The completing edges ww' and $v'w'$ are eligible by the orbit-cut exchange lemma. Localisation puts w, w' in C. Delete the opposite a-edge of S, insert ww', and complete $v'w'$ if missing by deleting the other selected ℓ-edge. All added edges lie in the new full square, so the square-block lemma permits no other newly created cycle. The old square is destroyed, and no other old cycle can contain an ℓ-edge: the first selected ℓ-edge is a bridge, while a cycle would require both selected edges of that colour.

If the parallel ℓ-edge was already selected, deleting the old square edge leaves a connected tree on C and inserting the new edge restores one cycle. Otherwise the deleted other ℓ-edge cannot lie outside C: after deleting the square edge, the two added internal edges would give the still connected C two cycles, although its only possible cycle is the new square. When the deleted edge is in C, its edge count remains $|C|$, and its only cycle is the new square; hence it is connected. All other components are untouched, proving the partition assertion. □

Fix an H-orbit $O \subseteq U$ and suppose some minimum selection has no tree component in O. Equivalently its number E_O of selected edges in O equals $|O|$. The following statements apply to every minimum selection having this equality.

Lemma 2 (Neighbours and middle transport inside a full orbit)**.** *Every full-graph neighbour of a vertex of a square S in a component $C \subseteq O$ lies in C. A global middle-colour move at S preserves $E_O = |O|$ and transports the component partition within O by its middle generator.*

Proof. Write the square colours as a, b. A full neighbour of colour a or b lies in S. If its colour k commutes with a, conjugating the selected a-edge by $\rho_k \in G_a$ gives an actual ambient-eligible a-edge. Lemma 1 puts both endpoints, including that neighbour, in C. The same argument applies when k commutes with b.

The only remaining case is $a = i-1, b = i+1, k = i$. Both square colours lie in I, so $i \in I$ and $\rho_i \in H$. Retain the four edges of S and transport every other selected edge by ρ_i, the accepted global middle move. This remains a minimum square-only selection. All H-orbits are preserved by ρ_i, and all four retained edges lie in O, so $E_O = |O|$ is preserved. Since every component of a minimum selection has at most one cycle, that equality forces every component within O to be unicyclic.

As an operation on Q^{ρ_i}, the move deletes S^{ρ_i} and inserts S. The deletion splits C^{ρ_i} into four trees. Every other component in O remains unicyclic. An inserted square vertex cannot lie in one of those other components, since it would then join two squares in one component. Hence all four inserted vertices lie in C^{ρ_i}. Inserting S reconnects all four pieces, since otherwise a tree in O would remain. Thus the partition within O is exactly the transported partition. Also $S \subseteq C^{\rho_i}$, equivalently $S^{\rho_i} \subseteq C$, proving the remaining neighbour assertion. Components with trees in other H-orbits cannot interfere: the middle generator preserves each orbit. □

Theorem 3 (A tree in every proper tail orbit)**.** *Every proper H-orbit $O \subsetneq \Omega$ in U has a tree component in a minimum-square tail selection. Consequently, if H is intransitive on Ω and has c nontrivial orbits, then*

$$2d \leq |U| - c. \tag{1}$$

Proof. Suppose O has no tree. Consider all minimum-square selections with $E_O = |O|$. By transitivity of the full G and properness of O, a full edge leaves O. Its inside endpoint lies in a selected component $C \subseteq O$, which has a unique square S. A selected path from S to that endpoint followed by the exit edge gives a connector. Minimize its length over all such selections, squares, selected paths and exit edges. The exit colour x satisfies $x \leq t$, since every generator in I preserves O. By Lemma 2 the selected path is nonempty.

Let its first edge be $v -_\ell w$, $v \in S$. If ℓ commutes with a square colour a, use the ordinary slide. Its other selected ℓ-edge cannot be farther along the connector's selected path. Indeed delete the first bridge and let W be the side containing w. Both selected a-edges lie outside W. If $w' = w^{\rho_a}$ were in W, its path from w would avoid a, contradicting eligibility. If it were in $C \setminus W$, its path from v^{ρ_a} would avoid ℓ, since both selected ℓ-edges are on or beyond the deleted bridge. This too contradicts eligibility. Thus this repeated-colour possibility would put w' outside C, contrary to Lemma 1. The slide therefore preserves the rest of the selected path, the exit edge (whose colour is outside I), and the orbit edge count. Its new square contains w, giving a shorter connector, a contradiction.

Otherwise S has colours $i-1, i+1$ and $\ell = i$. Here $i \geq t+2$, so ρ_i commutes with the exit generator ρ_x, since $x \leq t$. Perform the global middle move. Every selected edge of the connector after the first is outside S and therefore transports to a selected edge. The full exit edge transports to an actual x-edge. Its outside endpoint remains outside O, because $\rho_i \in H$ preserves O and its complement. The transported path begins at $w^{\rho_i} = v \in S$. Together with the transported exit it gives a connector shorter by one, still in a minimum selection with $E_O = |O|$. Possible vertex coincidences only shorten it further. This is the final contradiction.

Thus each proper nontrivial H-orbit contains a tree. Every tree component contributes one to vertices minus edges, and each unicyclic component contributes zero. Summing these deficits over the c nontrivial orbits gives (1). □

Corollary 4 (The intransitive-tail rank step)**.** *Suppose H has at least two nontrivial orbits on U and $t \leq (n - |U| - 1)/2$. Under the ambient-eligibility hypotheses above, $r \leq \lfloor (n-1)/2 \rfloor$.*

Proof. Equation (1) gives $2(r - 1 - t) \leq |U| - 2$, hence $2r \leq |U| + 2t \leq n - 1$. □

Source, attribution and remaining boundary. This is a proposed replacement for the *ambient-eligible* use of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.27, physical pages 34–36. Its literal statement assumes a paired fracture graph for H itself; that means crossing orbits of H_j, a weaker condition than crossing ambient G_j-orbits. The proof above does not identify these conditions. Later applications must supply two ambient-eligible edges of each tail colour, or take their separate unique-crossing branch. The broader literal local-eligibility assertion remains separate, not refuted by this distinction alone.

The fracture framework and rank target are the source's. The whole accepted paired-cut proof and equality-localisation proof remain paid dependencies. The global middle-colour move was supplied by the root collaborator and distinctly reviewed; the orbit-deficit argument here is a new proof adaptation of that accepted mechanism. No finite census, formal theorem, whole L044 recovery, original or question credit is asserted.